



Towards Operational Conversational Intelligence for Body-Worn Cameras

10 July 2026



A Speech Intelligence Framework for Robust Speaker-Attributed Conversation Analysis

Why Operational Conversational Intelligence for Body-Worn Cameras?

Current Challenges :

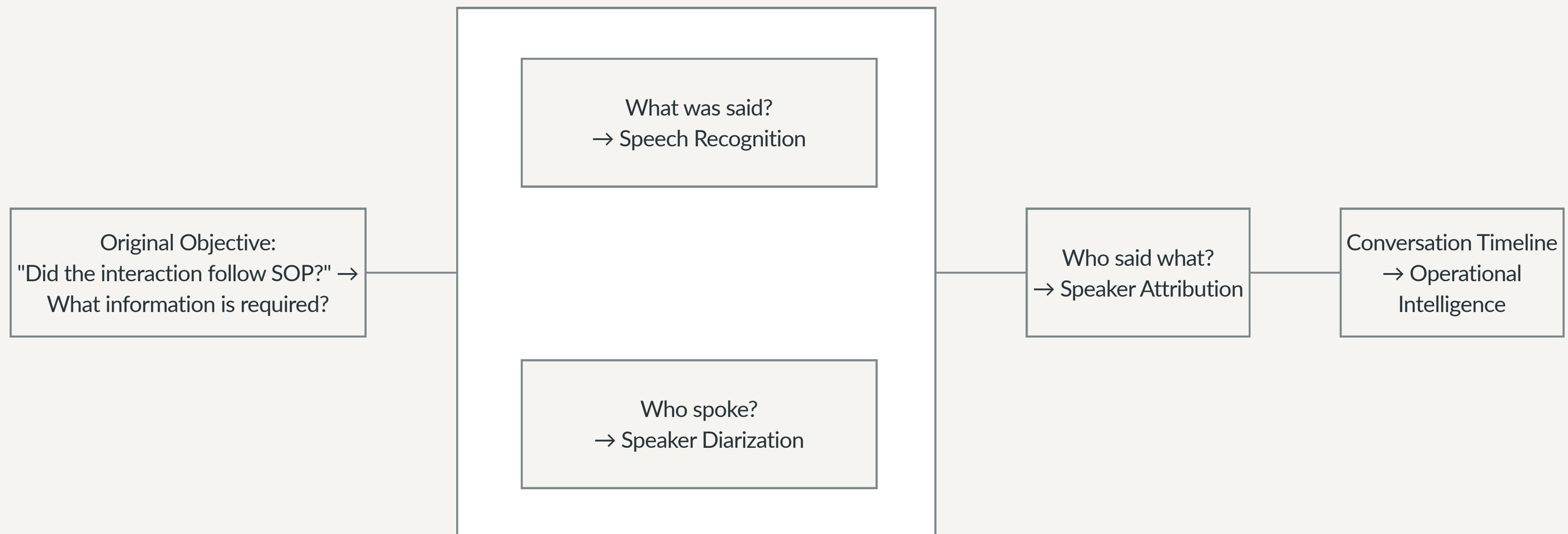
- Massive volumes of BWC recordings
- Manual review is slow & subjective
- Need operational understanding

BWC Audio is Fundamentally Different :

- Sirens & traffic noise
- Radio interference
- Wind & clipping
- Overlapping speech
- Moving speakers
- Emotional escalation



From Operational Intelligence to Speech Intelligence ■ ■ ■



Metric	VAD	ASR	Speaker Diarization
Source	Silero VAD	WhisperX Large-v3 + Wav2Vec2	Pyannote Diarization
Accuracy	0.7821	-	-
Precision	0.8250	-	-
Recall	0.8648	-	-
F1-Score	0.8445	-	-
Miss Rate	0.1352	-	0.1905 (19.05%)
False Alarm Rate	0.3971	-	0.1591 (15.91%)
Word Error Rate (WER)	-	0.2907 (29.07%)	-
Reference Words	-	485	-
Hypothesis Words	-	428	-
Substitutions	-	52	-
Insertions	-	16	-
Deletions	-	73	-
Diarization Error Rate (DER)	-	-	0.4599 (45.99%)
Speaker Confusion	-	-	0.1103 (11.03%)

consolidated
metrics table
extracted
exclusively for
the bwc video

Building the Speech Intelligence Foundation

Pre-Processing and VAD

Raw Audio



Acoustic Conditioning



Voice Activity Detection



Speech Grounding



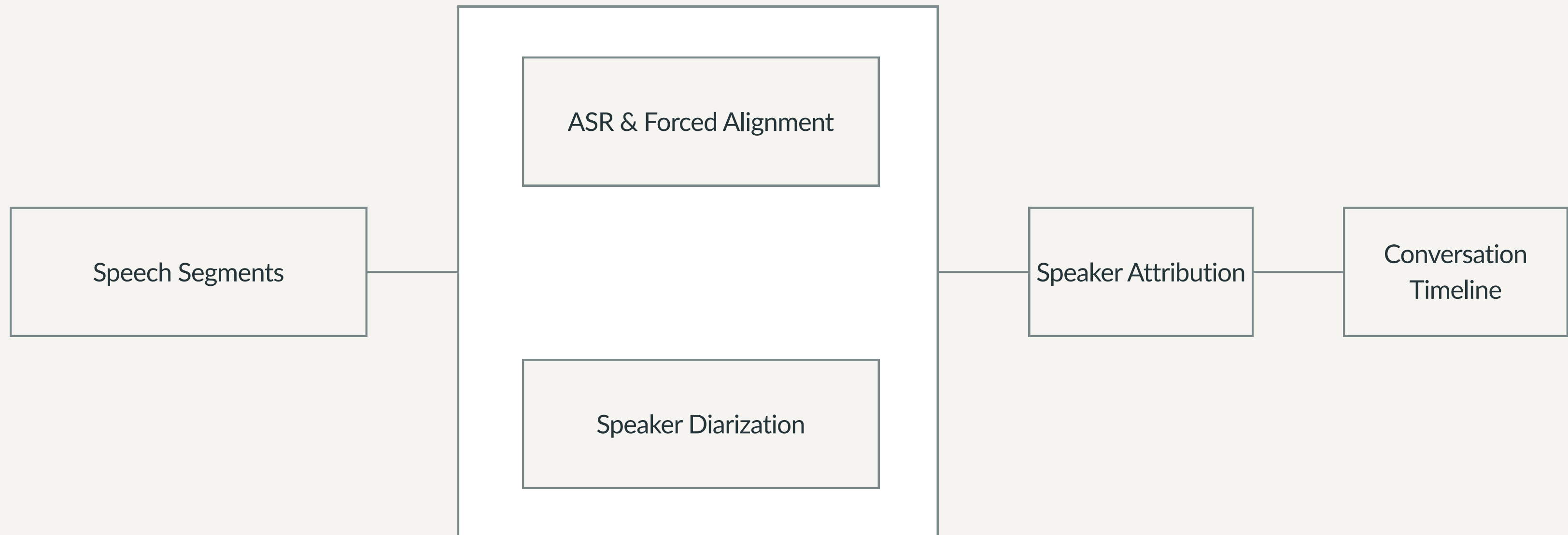
Speech Segments



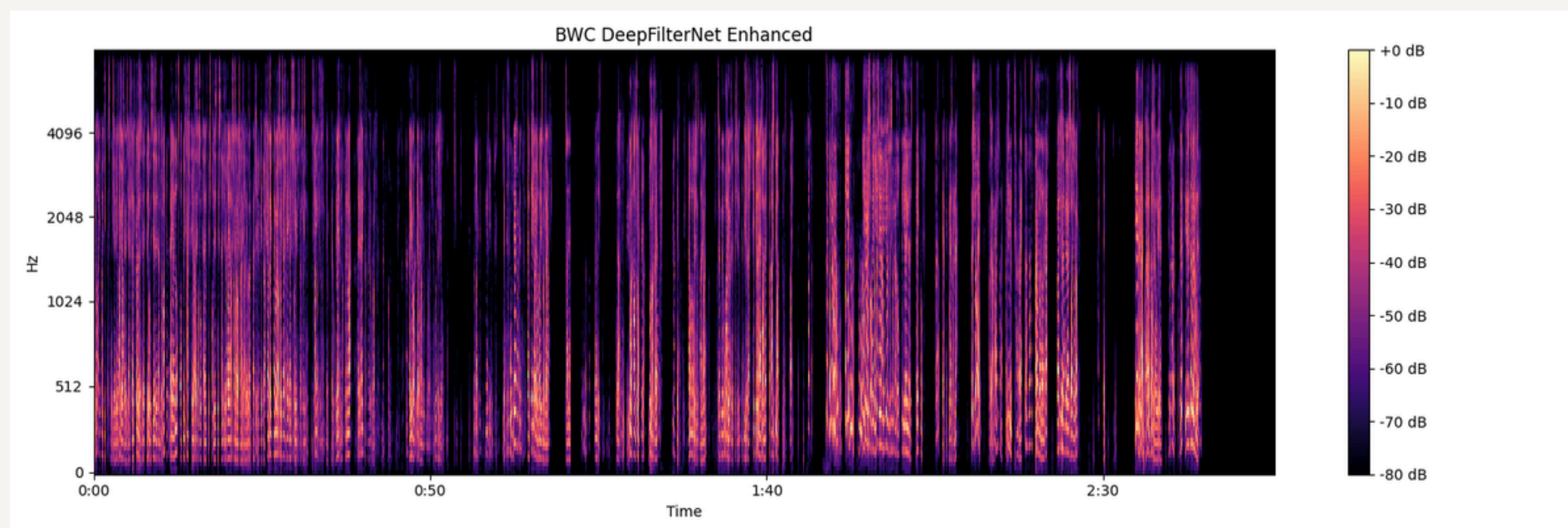
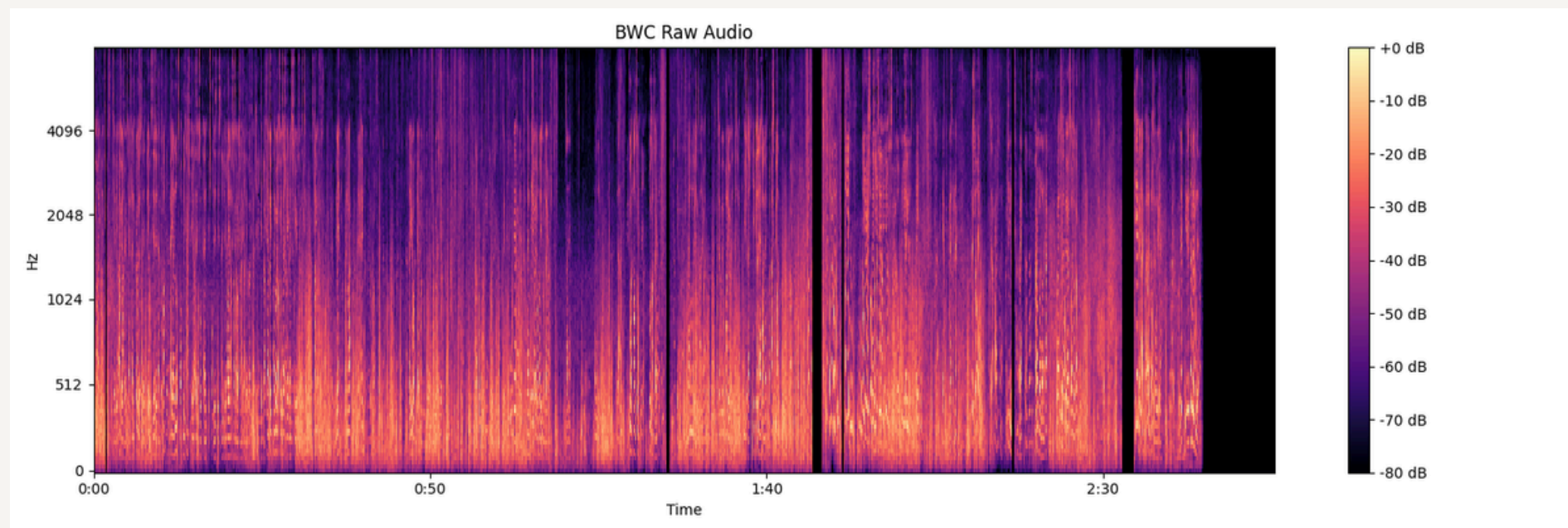
Key Findings

- ✓ High-recall VAD is critical for downstream pipeline performance
- ✓ RMS Normalization is safe and improves consistency
- ✓ Neural denoising degraded ASR — the Enhancement Trap

Transforming Speech into Conversations ■ ■ ■



Research Contributions



01

Framework Contributions

- End-to-End Speech Intelligence Framework for BWC audio
- Modular Pipeline enabling independent component evaluation

02

Scientific Finding: Enhancement Trap

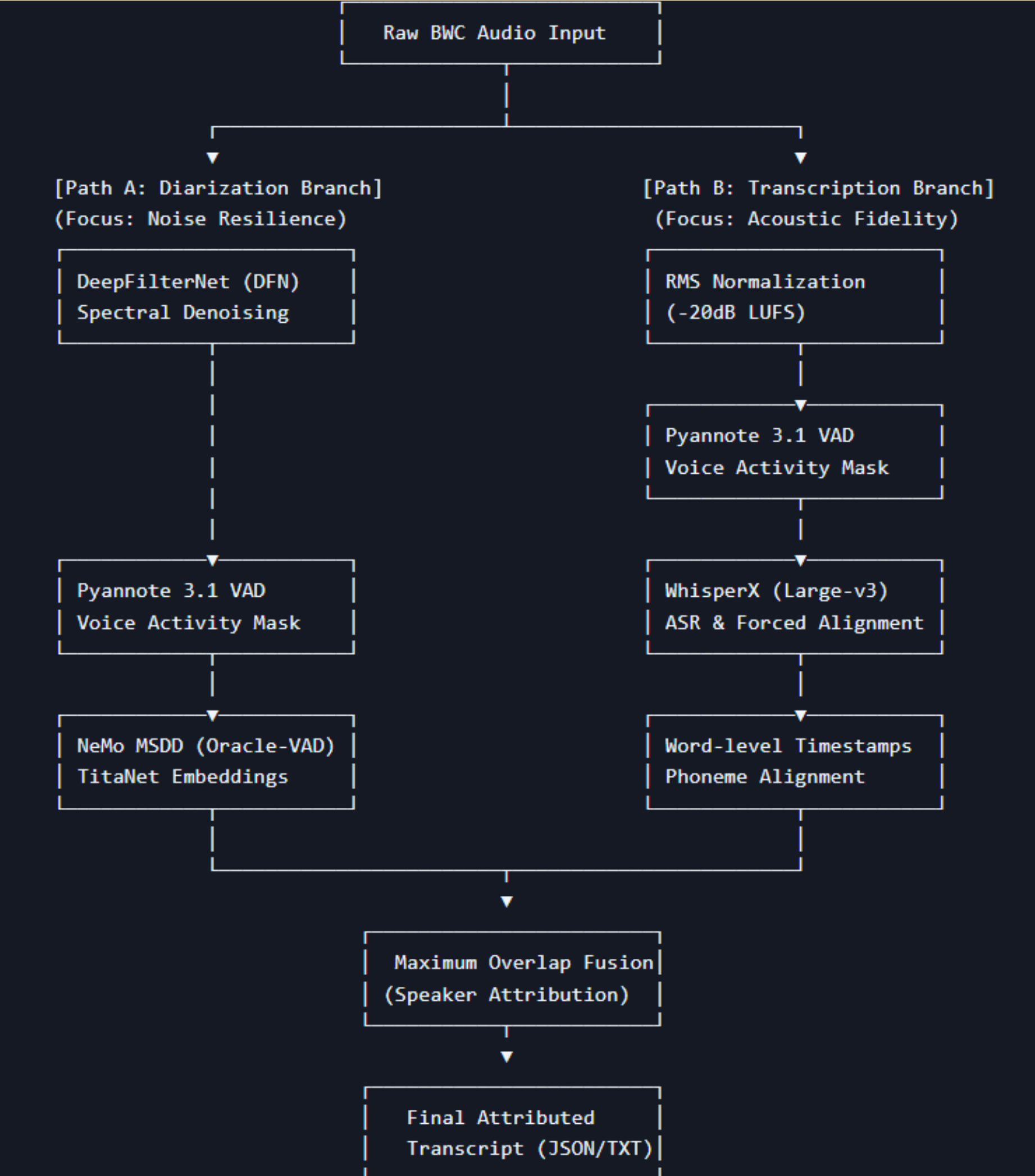
- Neural denoising shown to degrade downstream ASR performance in operational BWC audio conditions

03

Scientific Findings

- Oracle speech grounding provides reliable speech boundaries for speaker diarization
- VAD and Diarizer as Primary Bottleneck in speaker attribution

Final Speech
Intelligence Framework





VAD_DIR	
Metric	Value
Precision	0.8447
Recall	0.7721
F1 Score	0.8068
False Alarm	0.2990
Miss	0.2278

VAD_ASR	
Metric	Value
Precision	0.8416
Recall	0.7669
F1 Score	0.8025
False Alarm	0.3042
Miss	0.2330

ASR	
Metric	Value
WER	0.3243
S	84
I	27
D	46
Ref	484
Hyp	465

DIR	
Metric	Value
DER	0.4471
Miss	0.2300
False Alarm	0.1399
Confusion	0.0771

Results from
the final
pipeline we
showed in
previous slide



FUSION	
Metric	Value
WDER	0.09725
Speaker Accuracy	0.90274
Word Attribution Error	0.09725
Timeline Accuracy	0.7623

More than 75% accuracy
in timeline achieved.

This measures whether:

- speaker
- time
- word
- alignment

are simultaneously correct

Thank
You

